# Prathamesh Ettam

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## EDUCATION

**Centre for Development of Advanced Computing (C-DAC), Hyderabad,** 03/2024 – 08/2024
Post Graduate Diploma in Big Data Analytics(DBDA) Hyderabad, India

**WALCHAND INSTITUTE OF TECHNOLOGY (WIT), Solapur,** 06/2019 – 07/2023
BTECH, ELECTRONICS AND TELECOMMUNICATIONS 75.6%

## TECHNICAL SKILLS

- **Programming Languages & REST-API:** Python, Flask, FastAPI
- **Big Data Analytics**: Apache Kafka, Hadoop, ETL, PySpark, Amazon Kinesis, Amazon Redshift, Apache Airflow
- **Machine Learning**: Scikit-learn, Model Selection, Classification, Regression, Ensemble models
- **Deep Learning**: ANN, CNN, RNN, YOLO, Transfer Learning, TensorFlow, Computer Vision (OpenCV)
- **Data Science**: Statistics, Feature Engineering, EDA, Time Series, NLP, Hyperparameter Tuning
- **MLOps & DevOps**: MLflow, CI/CD, Docker, Jenkins, Kubernetes, Git, GitHub Actions
- **Generative AI**: LangChain, RAG, Vector Databases, Prompt Engineering, LLMs
- **AWS Tools**: AWS (Glue, EC2, S3, RDS, IAM, Elastic Beanstalk, CodePipeline, Athena, ECR, DynamoDB, EKS)
- **Databases**: PostgreSQL, SQL, NoSQL (MongoDB, Cassandra)
- **Data Analysis & Visualization**: Tableau, Amazon QuickSight, Pandas, Streamlit, Matplotlib, Seaborn

## PROJECTS

**Real-Time Voting System - Data Engineering**
- **Tech Stack:** Python, Kafka, Spark Streaming, PostgreSQL, Streamlit, Docker Compose.
- Designed and implemented a real-time voting system processing **1,000+** records per minute with **Kafka**.
- Devised and generated voter profiles with Python, ensuring real-time analytics for 1,000+ voters via Spark Streaming.
- Constructed a **Streamlit dashboard** for election stats, achieving a **5**-second response time.
- Secured **4,000+** voting records in PostgreSQL, ensuring data integrity and privacy.
- Orchestrated services with Docker Compose for scalable deployment in large elections.

**Subscription Renewal Prediction - MLOps**
- **Tech Stack**: Python, AWS (S3, EC2, ECR), MongoDB, Docker, GitHub Actions, CI/CD, FastAPI.
- Formulated a solution for subscription renewal forecasting, using **FastAPI** to deliver real-time model endpoints.
- Enhanced data quality by implementing validation, drift detection, preprocessing, and addressing class imbalance.
- Executed model training with **AdaBoost**, achieving **91%** accuracy, and produced artifacts and evaluation reports.
- Optimized deployment of the top model to AWS S3, achieving a **20%** boost in efficiency.
- Leveraged scalability with AWS EC2, orchestrated Docker via ECR, and automated **CI/CD** through GitHub Actions

**Real-time-Data-Ingestion-for-Smart City Streaming - Data Engineering**
- **Tech Stack:** IoT Simulation, Apache Kafka, Apache Spark, Docker, AWS (Glue, S3, Redshift, QuickSight, Athena).
- Created a comprehensive real-time **data pipeline** with simulated IoT data, boosting processing efficiency by **30%**.
- Synthesized data encompassing GPS coordinates, traffic cameras, weather conditions, and emergency incidents.
- Integrated Kafka for data ingestion and **PySpark** for processing, reducing latency by **40%** for quicker insights.
- Automated ETL with AWS Glue and Athena, boosting speed by **50%** and using S3 and Redshift for scalable analytics.
- Developed AWS QuickSight dashboards, accelerating decision-making by **25%** with ML-powered insights.

**CYBER-BOT - AI-Powered Cybersecurity Assistant - Generative AI (LLM)**
- **Tech Stack :**Python,Streamlit, SQLite, LangChain, HuggingFace, FAISS, GROQ API, Docker, gTTS, SpeechRecognition
- **Objective :** Crafted an advanced, functional, secure, and interactive chatbot for **CDAC Hyderabad**.
- Engineered a **95%** accurate chatbot with LangChain and HuggingFace, handling 100+ documents in diverse formats.
- Incorporated **RAG** to integrate external data sources with fresh knowledge, boosting response relevance by **20%** .
- Architected secure authentication, achieving **100%** success in password validation and data protection.
- Elevated user engagement by **50%** with **speech recognition** and **text-to-speech**, deploying with Docker and Streamlit

## CERTIFICATES

Machine Learning | Deep Learning | AWS | DevOps